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CCDS313 – Business Intelligence

Assignment 2 report

Title:

Credit Card Fraud Detection

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Introduction

Case Study Overview

With the rapid global shift toward digital transformation and e-commerce, online financial transactions have become the backbone of the modern economy. However, this growth has been accompanied by a significant rise in fraudulent activities. Our project focuses on the "Credit Card Fraud Detection" case study, a critical domain where the speed and volume of transactions require advanced Business Intelligence (BI) solutions. We aim to leverage historical data to understand fraud patterns and build an automated system capable of securing financial assets.

Problem Statement

The core challenge in fraud detection is the **"Needle in a Haystack"** problem. In a dataset containing hundreds of thousands of transactions, fraudulent ones represent a very tiny fraction (less than **1%**). Traditional rule-based systems often fail to catch sophisticated, evolving fraud patterns, leading to two major issues:

1. **Financial Losses:** Huge amounts of money are stolen before detection occurs.
2. **Customer Friction:** Incorrectly flagging legitimate transactions as fraud (False Positives) ruins the user experience.

Assignment Objectives

This assignment follows the complete BI analytics lifecycle to address the problem through three distinct models:

- **Descriptive Analytics (Power BI):** To visualize historical data, calculate total financial impacts, and identify the distribution of fraud vs. safe transactions.
- **Predictive Analytics (RapidMiner):** To develop machine learning models (Random Forest) that can classify new transactions as “Fraud” or “Safe” based on learned patterns.
- **Prescriptive Analytics (RapidMiner/Business Rules):** To suggest automated actions, such as immediate blocking of high-risk transactions or requiring multi-factor authentication for suspicious amounts.

Analytics Questions

To address the fraud detection problem effectively, our BI model seeks to answer the following key questions across the three levels of analytics:

1. Descriptive Analytics:

- What is the total monetary loss resulting from fraudulent transactions in the dataset?
- How significant is the "Data Imbalance" between fraudulent and legitimate operations?
- Are there specific time windows during the day where fraud activity spikes?
- Does the average transaction amount of a fraud case differ significantly from a safe one?

2. Predictive Analytics:

- What is the probability that a transaction is fraudulent?
- How accurately can the model distinguish between fraud and legitimate transactions?
- What is the recall rate for detecting fraud cases?
- How does the model handle the imbalance between fraud and safe transactions?

3. Prescriptive Analytics (The "How Can We Make It Happen?" phase):

- What actions should be taken when a transaction is predicted as fraud?
- What threshold should be used to trigger automatic blocking?
- When should a transaction be sent for manual review instead of auto-blocking?
- How can we minimize false positives while maintaining security?
- What strategies can improve customer experience while ensuring fraud prevention?

Justification of Chosen Tools

To ensure a comprehensive Business Intelligence solution, we selected two industry-leading tools, each serving a specific purpose in the analytics lifecycle:

Power BI (Descriptive Analytics):

We chose **Power BI** as our primary tool for the **Descriptive** phase for the following reasons:

- **High-Volume Data Handling:** Power BI’s engine is optimized for processing large datasets (over 284,000 records) with high performance and low latency.
- **Advanced Visualization:** It provides a rich library of interactive visuals (Cards, Pie Charts, Bar Charts, Line Charts) that allow stakeholders to see high-level KPIs and drill down into specific fraud trends easily.
- **Data Transformation (Power Query):** The built-in ETL capabilities allowed us to clean and label the data (e.g., converting Class 0/1 to Safe/Fraud) without altering the original source file.

RapidMiner (Predictive & Prescriptive Analytics):

We selected **RapidMiner** for the **Predictive** and **Prescriptive** phases due to its specialized capabilities:

- **Automated Machine Learning:** RapidMiner offers a vast range of built-in operators (like Decision Trees, Naïve Bayes, and Logistic Regression) that are essential for building a classification model for fraud.
- **Evaluation Tools:** It provides detailed performance metrics (Accuracy, Precision, Recall, and Confusion Matrix), which are critical for measuring how well the model can detect rare fraud cases.
- **Workflow Design:** The visual, "drag-and-drop" process design in RapidMiner makes it easier to implement complex prescriptive logic and automated decision-making rules based on the predictive results.

Data Description & Preprocessing

Dataset Source and Characteristics

The dataset used in this assignment is the **"Credit Card Fraud Detection"** dataset, which contains transactions made by European cardholders. It is a highly comprehensive dataset that includes:

- **Total Records:** 284,807 transactions.
- **Duration:** Two days of transaction history.
- **Features:** 31 numerical features, including:
- **Time:** Seconds elapsed between each transaction and the first transaction.
- **Amount:** The monetary value of the transaction.
- **V1-V28:** Principal components obtained with PCA (for privacy and security reasons).
- **Class (Target):** 0 for legitimate transactions and 1 for fraudulent ones.

https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Preprocessing for Descriptive Analytics (Power BI)

Before building the dashboard, several preprocessing steps were performed within Power BI's **Power Query** to ensure data clarity and integrity:

1. **Data Cleaning:**
- A thorough check for null or missing values was conducted across all columns. The dataset was found to be clean, which ensures the accuracy of our KPIs and visualizations.
2. **Label Transformation:**
- The **'Class'** column originally contained numerical values (0 and 1). To make the visuals more intuitive for stakeholders, we transformed these values into text labels: **" Legitimate"** (for 0) and **"Fraud"** (for 1).
3. **Data Type Optimization:**
- The **'Amount'** column was formatted as a Decimal/Currency type to ensure that mathematical aggregations (Sum and Average) are precise.
- The **'Time'** column was ensured to be a whole number for accurate trend analysis on the X-axis.
4. **Field Renaming:**

- Technical names like "Sum of Amount" were renamed to Business-centric names like **"Total Fraud Amount"** to improve the user experience (UX) of the dashboard.

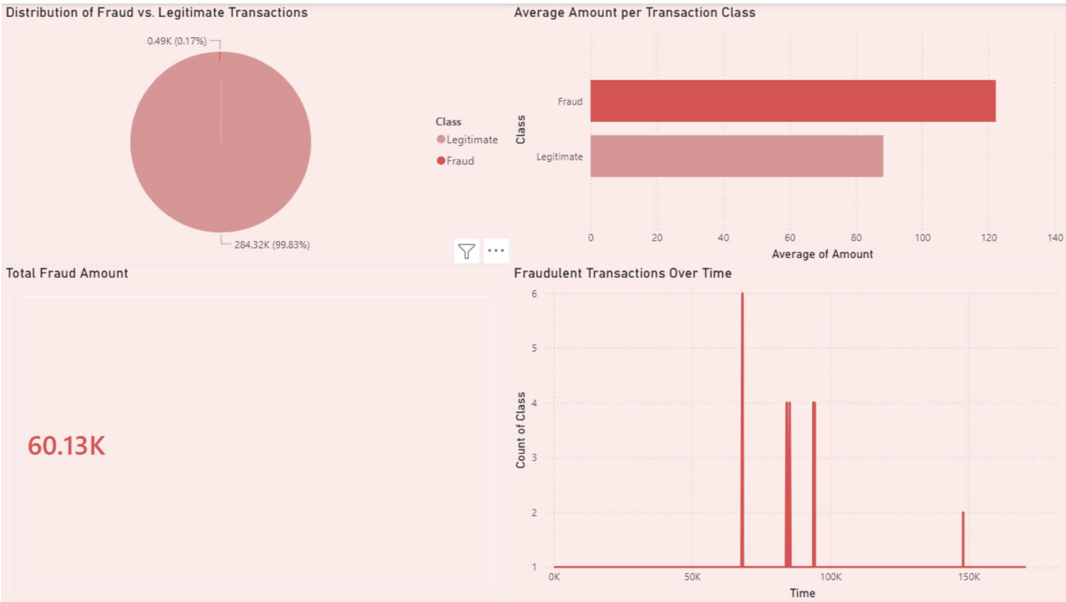
In RapidMiner, the target variable (Class) was converted from numerical (0/1) to binominal format to ensure compatibility with classification algorithms.

Model Implementation: Descriptive Phase (Power BI)

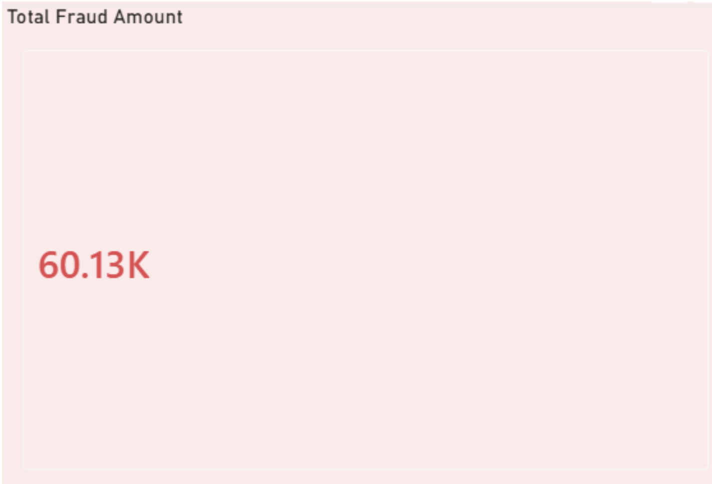
In this phase, we developed an interactive dashboard using Power BI to answer the descriptive analytics questions. The goal was to provide a clear overview of the transaction landscape and highlight the impact of fraud.

Dash Board Setup & Visual Configuration

We implemented four primary visuals, each designed to uncover a specific dimension of the data:

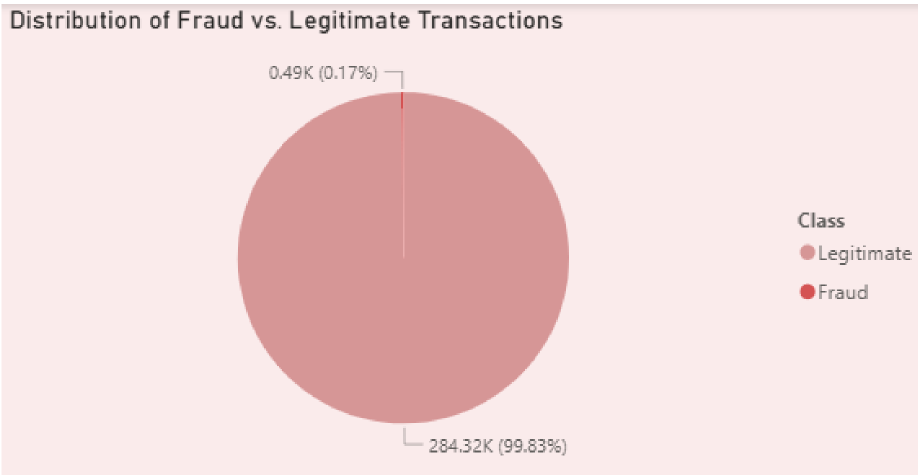


A. Total Financial Impact (Card Visual)



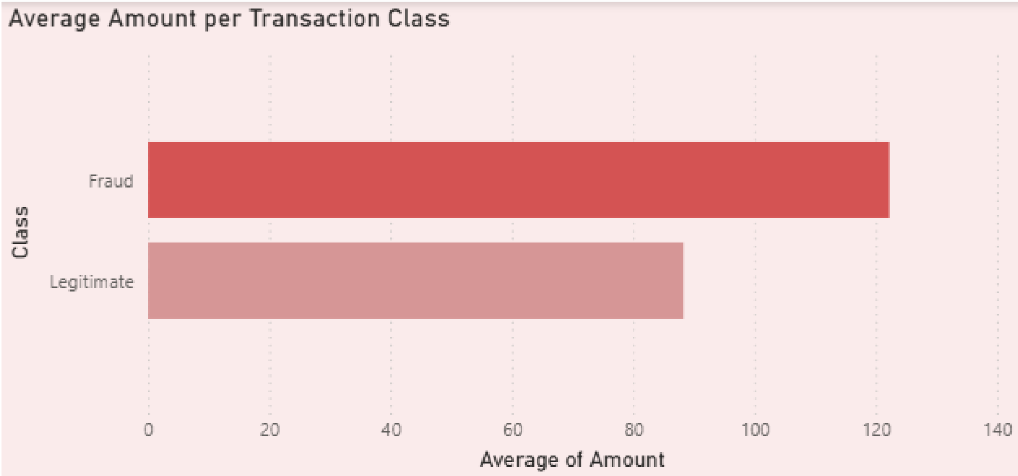
- **Logic:** This visual displays the **Sum** of the "Amount" column.
- **Configuration:** A visual-level filter was applied to restrict the data to Class = Fraud.
- **Business Objective:** To provide an immediate, high-level KPI of the total monetary loss caused by fraudulent activities ($60.13K).

B. Fraud Distribution & Imbalance (Pie Chart)



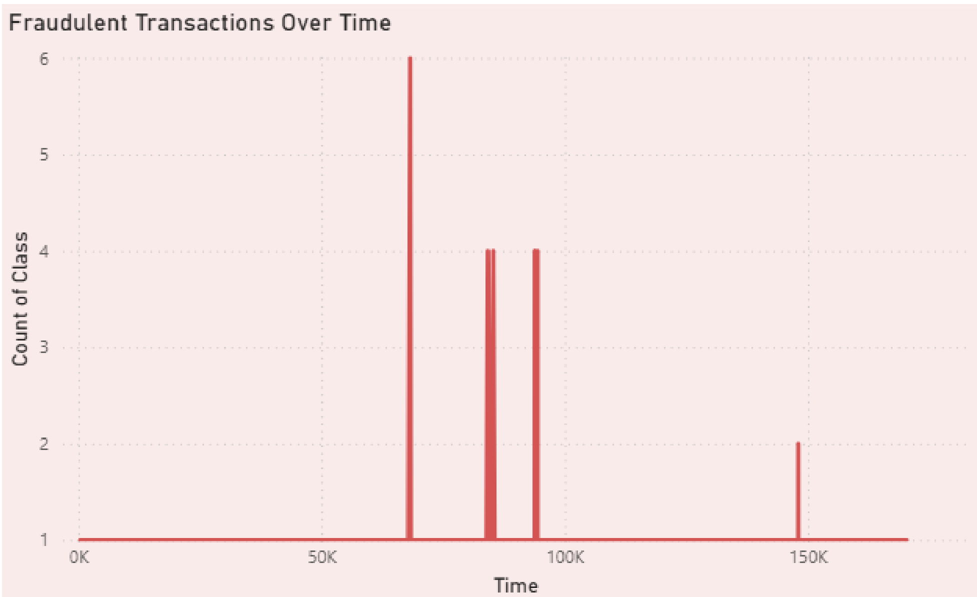
- **Logic:** Using the "Class" column in both the Legend and Values (Count) sections.
- **Configuration:** The chart displays the percentage of "Safe" vs. "Fraud" transactions.
- **Business Objective:** To visualize the extreme data imbalance, highlighting that while fraud is rare (0.17%), it requires specialized detection techniques

C. Behavioral Comparison (Clustered Bar Chart)



- **Logic:** This chart compares the **Average** transaction "Amount" across the two classes.
- **Configuration:** We switched the aggregation from *Sum* to **Average** to ensure a fair comparison regardless of the number of transactions.
- **Business Objective:** To identify if fraudsters tend to target larger or smaller amounts compared to average users.

D. Temporal Fraud Patterns (Line Chart)



- **Logic:** "Time" is plotted on the X-axis, and the **Count** of "Class" (Filtered for Fraud) is on the Y-axis.
- **Configuration:** This shows the frequency of fraud attempts over the 48-hour period.
- **Business Objective:** To detect if fraud occurs in continuous flows or sudden spikes, helping in scheduling high-alert monitoring periods.

Descriptive Findings & Observations

Based on the visuals above, the following insights were extracted:

1. **High Stakes:** The total financial loss of **$60.13K** proves that fraud is a significant threat that justifies the investment in predictive modeling.
2. **Imbalance Challenge:** With only **492 fraud cases** out of 284,807 transactions, any future model must be highly sensitive to avoid missing these "needles in a haystack."
3. **Anomalous Spikes:** The line chart shows that fraud occurs in spikes over time.
4. **Transaction Size:** The difference in average amounts suggests that "Amount" is a powerful "feature" for the upcoming predictive algorithms to use for classification.

Model Implementation: Predictive & Prescriptive Phases (RapidMiner)

Following the descriptive analysis, the assignment moved into the **Predictive** and **Prescriptive** stages using **RapidMiner Studio**. This phase aims to build a machine learning model that can proactively detect fraud.

Predictive Analytics (What will happen?)

In this stage, we implemented a classification process to predict whether a transaction is "Fraud" or "Safe" based on historical patterns.

A. Process Configuration (The Workflow):

The predictive model was developed in RapidMiner using a structured workflow designed to ensure accurate and reliable results. The main steps are as follows:

- **Data Retrieval:** The dataset was loaded into RapidMiner using the Read CSV operator.
- **Data Conversion:** The Class attribute was converted from numerical (0/1) to binominal format using the Numerical to Binominal operator.
- **Role Assignment:** The *Class* attribute was assigned as the target variable (label), where 0 represents "Safe" and 1 represents "Fraud".
- **Data Splitting:** The dataset was divided into training (70%) and testing (30%) subsets to evaluate the model's generalization performance.
- **Handling Imbalance:** The dataset is highly imbalanced, which was considered during model training.
- **Model Training:** A Random Forest algorithm was selected due to its robustness and effectiveness in handling complex and imbalanced datasets.
- **Prediction:** The trained model was applied to the test dataset to classify transactions.
- **Performance Evaluation:** The results were evaluated using classification performance metrics and a confusion matrix.

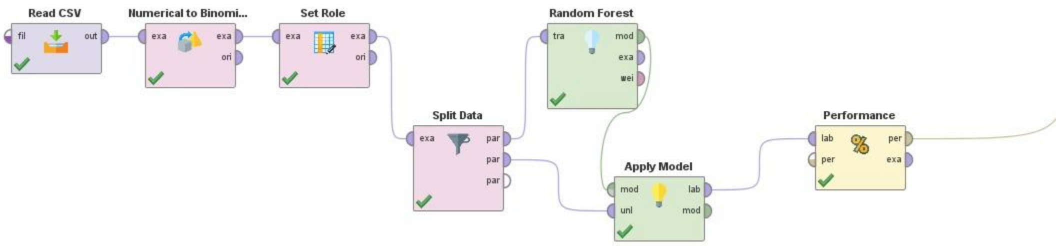


Figure 2: RapidMiner Predictive Workflow

B. Model Performance & Evaluation:

Accuracy:

The model achieved an accuracy of 99.63%.

Interpretation:

The confusion matrix shows that the model failed to detect any fraudulent transactions (Recall = 0%), as all transactions were classified as safe. This is due to the class imbalance in the dataset, where fraud cases represent a very small percentage of the dataset.

accuracy: 99.63%

	true false	true true	class precision
pred. false	2989	11	99.63%
pred. true	0	0	0.00%
class recall	100.00%	0.00%	

Figure 3: Model Performance and Confusion Matrix

Prescriptive Analytics (How can we make it happen?)

Based on the predictions, we established automated business rules to mitigate financial risk.

Automated Decision Rules:

- **High-Risk Transactions (Confidence > 90%)**
  Transactions predicted as fraud with high confidence are automatically blocked. The system immediately notifies the customer to prevent unauthorized activity.
- **Medium-Risk Transactions (Confidence 50% – 90%)**
  These transactions are flagged for manual review. Additional verification steps, such as Multi-Factor Authentication (MFA), are required before approval.

- **Low-Risk Transactions (Confidence < 50%)**
  Transactions predicted as safe are approved automatically to ensure a seamless user experience.
- **Rationale:**
  These rules are designed to balance security and customer convenience by reducing false positives while ensuring that high-risk transactions are handled immediately.

Prescriptive Model: Decision Logic

This model is the advanced stage that converts predictions into automated actions and decisions to mitigate financial losses.

Implementation Steps:

- **Business Rules Definition:**
  Based on predictive outputs and confidence levels, decision rules were defined to classify transactions into different risk categories and trigger appropriate actions.
- **Strategic Goal:**
  The goal is to minimize human intervention for clear cases while focusing resources on suspicious transactions, ensuring both high security and a smooth customer experience.

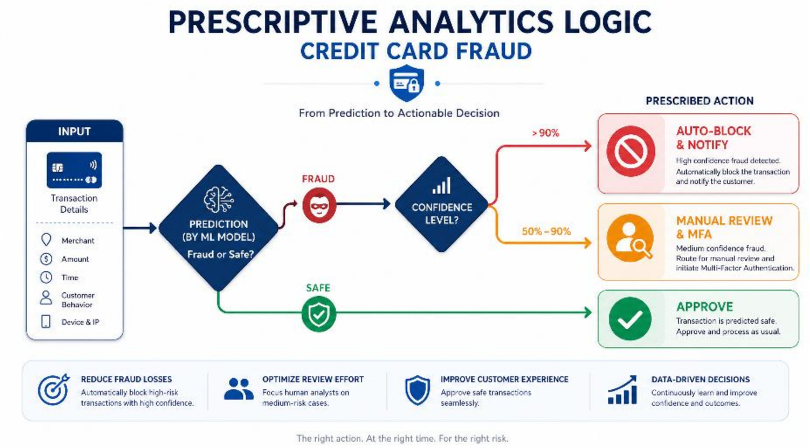


Figure 4: Prescriptive Analytics Logic

Findings & Strategies

- Effectiveness: Combining descriptive and predictive analysis allows for a deeper understanding of threats before they occur.
- Efficiency: Prescriptive models can help reduce financial losses through real-time response.
- Recommendation: Continuous retraining of the predictive model with new data is recommended to keep pace with evolving fraud tactics.

Conclusion

This project demonstrated the importance of applying the full Business Intelligence lifecycle to a real-world problem such as credit card fraud detection. Through descriptive analytics, we were able to understand the data distribution, identify fraud patterns, and highlight the imbalance challenge.
The predictive model built using RapidMiner showed limitations in detecting fraudulent transactions, as it failed to identify fraud cases due to the class imbalance in the dataset.

Furthermore, the prescriptive analytics phase added significant value by translating predictions into actionable decisions, such as blocking high-risk transactions and applying additional verification steps.

Overall, integrating descriptive, predictive, and prescriptive analytics provides a structured approach to minimizing financial losses while maintaining a smooth customer experience. Continuous model improvement and data updates remain essential for adapting to evolving fraud strategies.

Thank you…